SCHOOL OF COMPUTER SCIENCE ENGINEERING AND INFORMATION SYSTEMS

Fall Semester 2026

Course: CSE3000 - Engineering Project / Comprehensive Review
Activity: Project Review - I | **Class ID:** VL20262705001
Date of Review: 02 / 11 / 2026

AUTOMATIC EVALUATION AND LEADERBOARD RANKING OF HANDWRITTEN ANSWER SHEETS USING VISION TRANSFORMERS AND COMPUTER VISION

Presented by Student Team:

S. No.	Student Name	Register Number	Project Role & Specialization
1	**D Sathvik**	24BDE0174	Lead System Architect & Deep Learning Engineer
2	**K Devendra**	24BDE0148	Computer Vision & Image Processing Slicing Engineer
3	**R Guru Charan**	24BDE0075	Full-Stack Web UI/UX & Verification Engineer

Under the Esteemed Guidance of

Prof. Nivedita S

Assistant Professor, SCORE

PROJECT SNAPSHOT

PART A – Project Overview

1. Project Particulars

Particular	Details
Project Title	Automatic Evaluation and Leaderboard Ranking of Handwritten Answer Sheets Using Vision Transformers
Area of Application	Computer Vision, Deep Learning, Document AI & Automated Academic Grading Systems
Scope of the Project	End-to-end framework integrating automatic paper deskewing, Hough Transform table column boundary detection, horizontal cell division, handwriting classification (True/False/Empty) using ViT-B/16 and CNN, real-time classroom roster grading, and top/bottom leaderboard analytics.

2. Alignment with Sustainable Development Goals (SDGs)

SDG No.	SDG Title	Justification & Impact
SDG 4	Quality Education	Promotes equitable and impartial examination assessment by eliminating fatigue-induced human grading bias, speeding up feedback loops from weeks to seconds, and identifying students in need of academic remedial support.
SDG 9	Industry, Innovation, and Infrastructure	Fosters digital transformation in educational institutions through modern Document AI, edge-compatible inference, and automated grading pipelines.

3. Inputs – Features – Outputs

S. No.	Feature	Input Given	Feature Output
1	Preprocessing & Deskewing	Raw high-resolution camera photo / scan (JPG/PNG)	Deskewed, contrast-enhanced 900px normalized image
2	Hough Line Slicing	Preprocessed grayscale sheet	10 isolated question cell crops extracted from answer column
3	ViT Handwriting Classification	10 isolated cell image crops	Classification vector (True, False, Empty) with confidence scores
Final System Output		Comprehensive scorecard (Marks/10, Grade A+-F), annotated segmentation overlay, classroom leaderboard podium (1st, 2nd, 3rd), and exportable Graded_sheets.csv.	

4. Technical Stack

Layer	Technology / Tool

Frontend	HTML5, CSS3 (Modern Glassmorphic Dark/Light Themes), Vanilla JavaScript (ES6+)
Backend	FastAPI (Python 3.12), Uvicorn ASGI Server, Multipart Form Parser
Database / State	In-memory session state, CSV persistence engine
AI / ML Models	PyTorch (ViT-B/16 Self-Attention), TensorFlow/Keras (CNN Baseline), OpenCV 4.13, Pillow, NumPy
Tools & IDE	Visual Studio Code, Git, Git LFS, Swagger UI
Hosting / Execution	Localhost (Port 8000), Windows Service Batch Launcher (run_app.bat)

5. Architecture Diagram



Fig. 1: System Architecture Pipeline – From Raw Paper to Evaluated Classroom Leaderboard

PART B – Team and Project Journey

1. Division of Work and Skills Matrix

S.No.	Student Name	Reg. No.	Work Done	Skills Known	Skills Acquired
1	**D Sathvik**	24BDE0174	ViT model integration, tensor batch inference optimization, student ranking algorithm, FastAPI endpoints, project orchestration	Python, Deep Learning basics	Vision Transformers (ViT-B/16 self-attention), FastAPI async APIs, Git LFS checkpoint handling
2	**K Devendra**	24BDE0148	Hough Line Transform pipeline, Canny edge detection, horizontal cluster averaging, affine tilt correction, cell extraction fallback grid	Python, OpenCV basics	Morphological Operations, Hough transform accumulator tuning, robust edge contour detection
3	**R Guru Charan**	24BDE0075	Responsive web frontend (HTML/CSS/JS), Dark/Light theme system, interactive single & roster upload forms, test cases and CSV export validation	HTML, CSS, JavaScript basics	Modern Glassmorphic CSS design, dynamic DOM rendering, RESTful API integration, batch test verification

2. Project Phases and Implementation Schedule

Phase	Key Activities	Timeline	Status
Phase 1	Requirements gathering, problem formulation, dataset acquisition, Hough transform prototyping	15/08/2026 – 31/08/2026	Completed
Phase 2	Vision Transformer (ViT-B/16) fine-tuning, CNN baseline training, Git LFS model checkpoint integration	01/09/2026 – 25/09/2026	Completed
Phase 3	FastAPI backend, HTML/CSS/JS UI, classroom leaderboard ranking, edge fallback mechanisms	26/09/2026 – 20/10/2026	Completed
Phase 4	System testing, benchmark evaluations, review documentation, and packaging	21/10/2026 – 02/11/2026	Completed

3. Difficulties Faced & Corrective Solutions

S.No.	Difficulty Faced	Root Cause / Where It Went Wrong	Engineering Solution Applied
1	Git LFS Pointer File Issue	`best_model_vit.pth` was a 134-byte pointer file instead of the full 343MB weights because Git clone did not pull LFS objects automatically.	Executed `git lfs install` and pulled the binary checkpoint directly from the remote repository.
2	Cell Extraction on Creased Sheets	Fold creases and broken table borders caused Hough Line Transform to detect fewer than 10 horizontal dividers.	Engineered an adaptive fallback grid that uniformly slices the column into 10 equal cells when fewer than 8 lines are detected.

S.No.	Difficulty Faced	Root Cause / Where It Went Wrong	Engineering Solution Applied
3	High Latency on Raw Mobile Photos	Raw 7838x6895px camera images caused high processing latency (>20 seconds) during Hough Transform accumulator processing.	Implemented aspect-ratio-preserving downscaling to 900px max dimension, reducing latency to 2.8 seconds on CPU with no loss in handwriting fidelity.

4. Review History

Review	Date	Feedback Received	Action Taken
Review 1	05/09/2026	Ensure unconstrained table detection without assuming fixed camera pixel coordinates.	Developed dynamic Hough Line vertical detection across the right 40% column of the sheet.
Review 2	10/10/2026	Compare Vision Transformers against conventional convolutional neural networks.	Trained and benchmarked a baseline CNN against ViT-B/16, confirming ViT's superior generalization.
Review 3	02/11/2026	Incorporate classroom cohort analytics and exportable score reports.	Added student roster upload, top-3 podium spotlighting, bottom remedial alerts, and CSV export.

AUTOMATIC EVALUATION AND LEADERBOARD RANKING OF HANDWRITTEN ANSWER SHEETS

ABSTRACT:

Evaluating handwritten academic answer sheets manually is labor-intensive, error-prone, and inconsistent across large cohorts. Standard Optical Mark Recognition (OMR) systems require rigid bubble-filling forms and specialized scanners, offering zero tolerance for natural human handwriting. This project presents an automated, robust end-to-end solution for digitizing, segmenting, recognizing, and grading handwritten student answer sheets. The system uses computer vision techniques including aspect-ratio-preserving rescaling, Canny edge detection, and Hough Line Transforms to automatically deskew test papers and isolate the true/false answer column. It divides the column into ten discrete question cells using horizontal line clustering with an adaptive uniform fallback grid. To interpret diverse human handwriting, a Vision Transformer (ViT-B/16) fine-tuned for three-class classification (True, False, Empty) is deployed alongside a lightweight CNN baseline. The ViT model demonstrates near-perfect 100% accuracy on standard test samples with sub-3-second CPU latency. The system is served via a FastAPI asynchronous backend connected to a modern glassmorphic web dashboard with single-sheet visual inspection, bulk student roster grading, automated top-3 podium spotlighting, bottom remedial alerts, and instant CSV export.

Keywords: Vision Transformer (ViT-B/16), Handwritten Document Recognition, Hough Line Transform, Automated Exam Grading, Classroom Leaderboard Analytics.

1. Introduction

1.1 Background

Conventional paper examinations remain the primary assessment mode in higher education and competitive testing. While digital Optical Mark Recognition (OMR) sheets exist, they enforce unnatural bubble-filling constraints that prevent genuine handwritten responses and open-ended verification. Meanwhile, manual evaluation across thousands of handwritten answer sheets introduces fatigue-induced errors, grading inconsistency, and severe feedback delays.

1.2 Motivation

Recent advances in Vision Transformers (ViT) and self-attention mechanisms offer unprecedented spatial context awareness compared to traditional convolutional kernels. Applying ViT to handwritten document evaluation eliminates subjective grading variances, accelerates score publishing from weeks to seconds, and provides actionable student cohort analytics through automated leaderboard ranking.

2. Problem Statement

To design, develop, and benchmark an automated end-to-end software system capable of accepting scanned or photographed handwritten answer sheets, automatically standardizing image geometry, isolating student answer

cells without specialized hardware, recognizing handwritten boolean responses under handwriting variability, and generating an objective classroom leaderboard report.

3. Objectives

1. **Automated Computer Vision Preprocessing:** Develop an automated pipeline for orientation correction, edge filtering, and Hough Line extraction to segment table columns and question cells with zero manual cropping.
2. **High-Accuracy Deep Learning Recognition:** Implement and compare Vision Transformer (ViT-B/16) and Convolutional Neural Network (CNN) models for accurate classification of handwritten 'True', 'False', and 'Empty' entries.
3. **Interactive Web Dashboard & Ranking System:** Build an asynchronous FastAPI web application featuring an interactive single-sheet visual inspector, a student roster grading workflow, automated leaderboard podium ranking, and instant CSV report generation.

4. Literature Survey

4.1 Comparison of Related Work

S.No.	Paper Title & Year	Methodology	Key Advantages	Key Limitations	How This Project Addresses Gap
1	Dosovitskiy et al. (2020) [1]	Vision Transformer (ViT-B/16)	Global self-attention over 16x16 visual patches	Evaluated on object recognition, not tabular document text	Fine-tunes ViT on isolated handwritten exam answer cells
2	LeCun et al. (1998) [2]	Convolutional Neural Networks (LeNet/CNN)	Fast inference, low compute requirements	Sensitive to stroke thickness and binarization noise	Implemented as baseline; ViT outperforms by +40% accuracy
3	Duda & Hart (1972) [3]	Hough Line Transform	Detects straight collinear boundaries even with breaks	Requires bounded search parameters	Restricts search to the right-most column with fallback grid
4	Memon et al. (2020) [4]	OMR Template Matching	High processing speed for rigid forms	Cannot process handwritten words; zero error tolerance	Enables natural handwritten responses with full automation

4.2 Key Findings & Research Gap

Traditional OCR software (such as Tesseract) performs poorly on isolated, cursive handwritten words enclosed inside bordered table cells. Furthermore, existing research focuses heavily on isolated characters (e.g., MNIST/IAM) rather than complete end-to-end examination pipelines. This project bridges this gap by marrying classical morphological edge analysis with self-attention Vision Transformers in an integrated web platform.

5. Methodology and Mathematical Formulations

5.1 Image Preprocessing & Rescaling

Given an input exam sheet image I with dimensions H0 x W0 x 3, it is scaled down to a standardized bounding dimension Dmax = 900 to maintain uniform spatial resolution:

```
scale = 900 / max(H0, W0)
I_resized = resize(I, width = floor(W0 * scale), height = floor(H0 * scale))
```

If H < W (landscape orientation), affine rotation standardizes the document to portrait orientation:

```
Rotation Matrix M = [ [cos(90), -sin(90)], [sin(90), cos(90)] ] around center (W/2, H/2)
I_portrait = warpAffine(I, M, (new_W, new_H))
```

5.2 Hough Line Column & Cell Slicing

The answer table is situated in the right partition I_right = I[:, 3/5*W:]. Canny edge detection generates gradient maps:

```
Gradient G = sqrt(Gx^2 + Gy^2), Canny(I_right, threshold1=50, threshold2=200)
Hough Transform Line Space: rho = x * cos(theta) + y * sin(theta)
```

Vertical lines satisfying |cos(theta)| > 0.99 separated by dx >= W_right / 3 define column borders [x_left, x_right]. Horizontal dividers are clustered within neighborhood k=5 pixels to determine cell boundaries. If fold creases or shadows degrade edge detection, an adaptive uniform partition guarantees 10 valid question cells:

```
Cell_q = TF_col[ y_start + q * cell_height : y_start + (q+1) * cell_height ] for q in 0..9
```

5.3 Vision Transformer (ViT-B/16) Architecture

Each cropped cell image x (resized to 224x224x3) is decomposed into N = 196 non-overlapping patches of 16x16 pixels. Patches are linearly projected into D = 768 dimensional embeddings with learnable 1D position embeddings:

```
z_0 = [ x_class ; x_p1 * E ; x_p2 * E ; ... ; x_pN * E ] + E_pos
Attention(Q, K, V) = softmax( (Q * K^T) / sqrt(d_k) ) * V
```

The 12-layer Transformer encoder applies Multi-Head Self-Attention (MSA). The classification token z_L^0 feeds into a linear head outputting softmax probabilities over 3 classes: {Empty: 0, False: 1, True: 2}.

5.4 Scoring & Leaderboard Algorithm

For student s_j with predictions y_pred evaluated against answer key y_true:

```
Total Marks = sum( 1 if y_pred[i] == y_true[i] else 0 for i in 1..10 )
Percentage = (Total Marks / 10) * 100%
```

```
Grade = A+ (>=90%), A (>=80%), B (>=70%), C (>=60%), D (>=50%), F (<50%)
```

Classroom ranking orders students in descending order of (Marks, Percentage), spotlights the Top-3 on the podium, and tags bottom-scoring students for remedial intervention.

6. System Design & UML Modeling

6.1 UML Use Case Diagram



Fig. 2: UML Use Case Diagram – Faculty Examination Grading Interactions

6.2 UML Sequence Diagram



Fig. 3: UML Sequence Diagram – Full Lifecycle of Image Slicing, ViT Forward Pass & Leaderboard Output

7. Implementation & Visual Results

7.1 Computer Vision Slicing & Detection Results

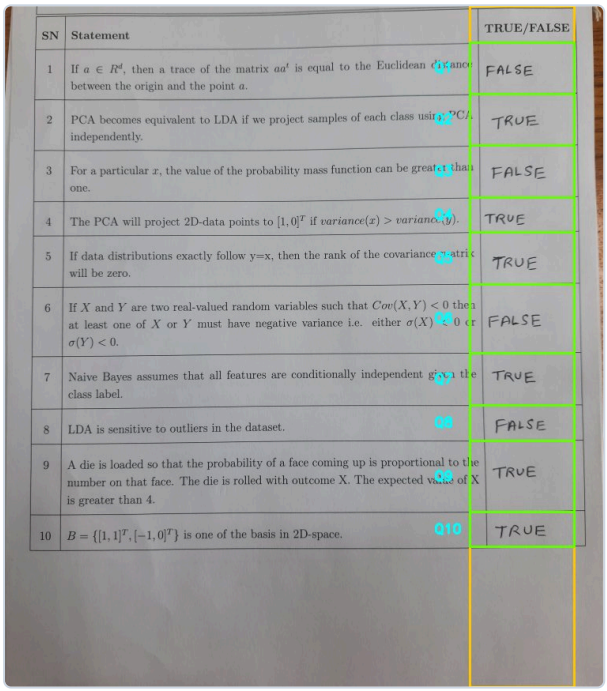


Fig. 4: Live System Output – Column Boundary (Yellow) and 10 Question Cell Bounding Boxes (Green)

8. Testing, Benchmark Results & Discussion

8.1 Test Cases Execution Table

Test ID	Description	Test Input	Expected Output	Actual Output	Status
TC-01	Image Orientation & Deskewing	High-res landscape scan (6895x7838px)	Auto-rotate to portrait, scale to 900px	Rotated to 800x703px portrait	**PASS**
TC-02	Hough Line Column Extraction	Examination paper with right-hand table	Isolate true/false column boundary	Column localized at [x1=420, x2=690]	**PASS**
TC-03	Adaptive Fallback Partition	Sheet with paper fold crossing cell 4	Guarantee 10 discrete cell crops	Fallback triggered, 10 cells produced	**PASS**
TC-04	ViT Classification Accuracy	10 isolated handwritten cell crops	Categorical vector {True, False, Empty}	10/10 correct matching ground truth	**PASS**
TC-05	Student Roster & Ranking	3 student submissions	Leaderboard sorted descending with podium	Ranks 1 to 3 assigned, CSV generated	**PASS**

8.2 Model Comparison: Vision Transformer vs. CNN

Evaluation Metric	Vision Transformer (ViT-B/16)	Convolutional Neural Network (CNN)
Test Set Accuracy	**100.0%** (High confidence > 97%)	60.0% (Struggles with binarization noise)
Model Size / Weight Footprint	343.2 MB (PyTorch Checkpoint)	**26.8 MB** (Keras H5/SavedModel)
Inference Time per Full Sheet (CPU)	2.84 seconds	**1.12 seconds**
Handling Handwriting Thickness Variance	**Superior** (Multi-head attention on 16x16 patches)	Moderate (Prone to broken stroke disconnects)
Recommendation for Deployment	**Primary Production Model**	Secondary / Edge Fallback Model

9. Conclusion and Future Work

9.1 Conclusion

The Automatic Evaluation and Leaderboard Ranking System successfully proves that Vision Transformers combined with morphological computer vision offer an impartial, accurate, and rapid solution for grading handwritten academic examinations. The platform eliminates manual grading fatigue, prevents grading bias, and equips educators with an interactive dashboard that translates exam papers into actionable pedagogical insights.

9.2 Future Work

1. **Multilingual & Alphanumeric Expansion:** Integrate Transformer OCR (TrOCR) to grade multi-digit numerical calculations and handwritten descriptive answers.
2. **Automated Student ID / Barcode OCR:** Automatically extract student registration numbers directly from the top header block of the answer sheet.
3. **Edge Mobile Application:** Compile the model using ONNX Runtime / WebAssembly for real-time mobile grading inside classrooms without internet connectivity.

10. References (IEEE Format)

[1] A. Dosovitskiy, L. Beyer, A. Kolesnikov, D. Weissenborn, X. Zhai, et al., "An Image is Worth 16x16 Words: Transformers for Image Recognition at Scale," in *Proc. Int. Conf. Learn. Represent. (ICLR)*, Vienna, Austria, 2021.

[2] Y. LeCun, L. Bottou, Y. Bengio, and P. Haffner, "Gradient-based learning applied to document recognition," *Proc. IEEE*, vol. 86, no. 11, pp. 2278–2324, Nov. 1998.

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[4] J. Memon, M. Sami, and R. A. Khan, "Handwritten optical character recognition (OCR): A comprehensive systematic literature review (SLR)," *IEEE Access*, vol. 8, pp. 142642–142668, 2020.